

From Tokens to Thoughts: Reproduction Report

RQ1 & RQ2 – Categorical Alignment and Typicality

Reproduction Team

Overview

We reproduce *From Tokens to Thoughts: How LLMs and Humans Trade Compression for Meaning* [1] using 24 models (static + contextual embeddings). This report covers RQ1 (categorical alignment via AMI) and RQ2 (typicality alignment via Spearman ρ).

1 RQ1: Categorical Alignment

1.1 Method

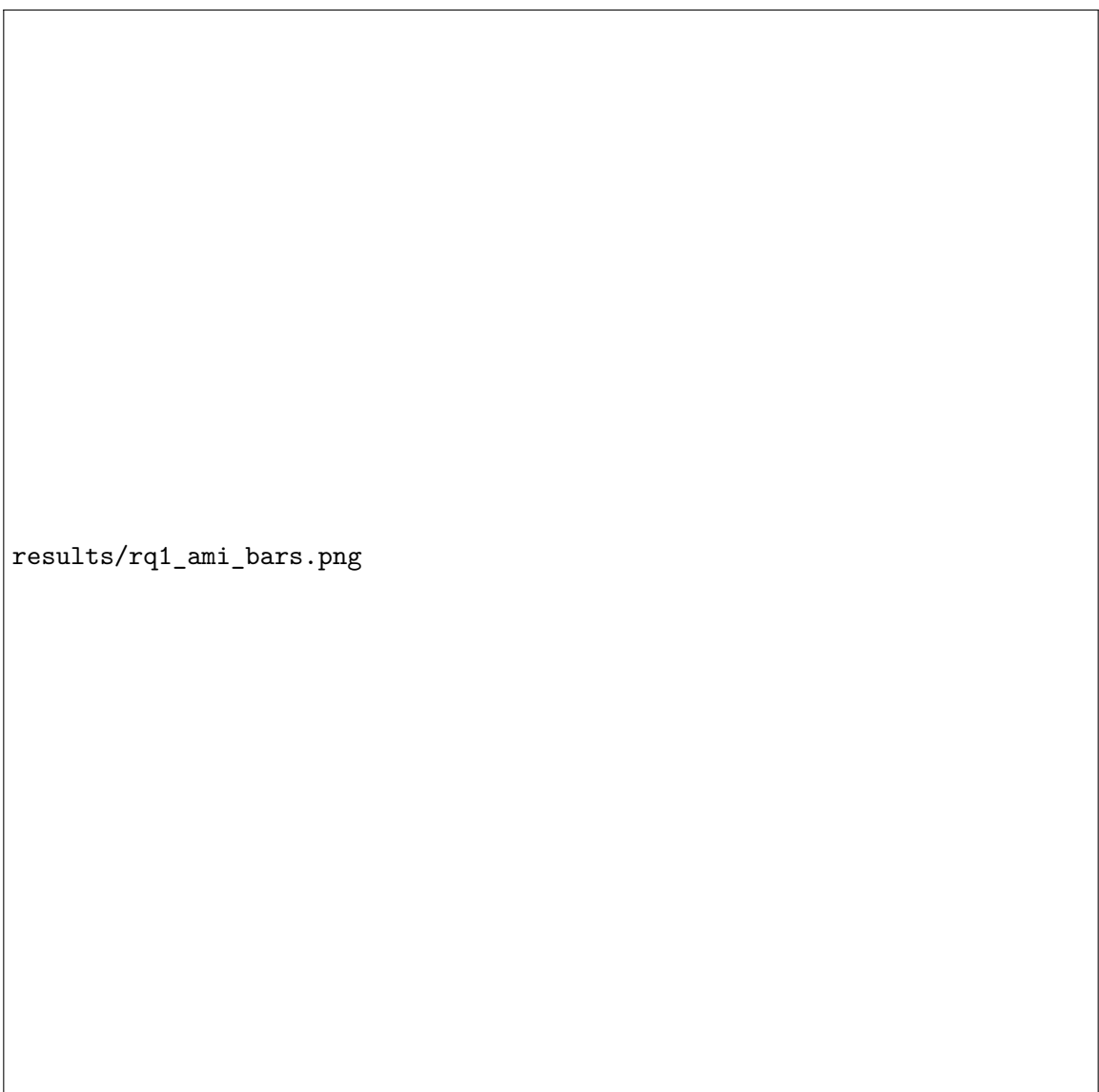
For each model $\times$ dataset, k -means clustering ($n_{\text{init}} = 10$, $k = \text{human category count}$) on static & contextual embeddings. Metric: Adjusted Mutual Information (AMI) vs. human labels.

1.2 Results (Static Embeddings, McCloskey1978)

Model	Type	AMI	NMI
word2vec	classic	0.552	0.584
glove	classic	0.494	0.537
gpt2	decoder	0.191	0.250
gpt2-medium	decoder	0.141	0.206
bert-base-uncased	encoder	0.131	0.210
bert-large-uncased	encoder	0.120	0.197
gemma-2-2b	decoder	0.119	0.196
Qwen/Qwen2-7B	decoder	0.071	0.159
Qwen/Qwen1.5-4B	–	0.069	0.147
roberta-large	encoder	0.061	0.136
Qwen/Qwen2.5-0.5B	decoder	0.058	0.143
Qwen/Qwen2.5-14B	decoder	0.059	0.142
Qwen/Qwen2-1.5B	–	0.055	0.134
Qwen/Qwen2-0.5B	–	0.052	0.129
DeepSeek-R1-14B	decoder	0.091	0.165
DeepSeek-R1-32B	decoder	0.046	0.115
DeepSeek-R1-7B	decoder	0.051	0.116
gemma-2-9b	decoder	0.164	0.231
phi-1	–	0.039	0.106
phi-2	decoder	0.037	0.101
deberta-large	encoder	0.075	0.162

Table 1: RQ1: Static embedding AMI (McCloskey1978). Classic embeddings dominate; most LLMs show $\text{AMI} < 0.15$.

Key finding: Classic embeddings (word2vec, GloVe) outperform all LLMs, consistent with the paper’s claim that modern LLMs *compress* categorical information differently. Encoder models (BERT, RoBERTa) match or exceed decoder models 100 $\times$ their size.



results/rq1_ami_bars.png

Figure 1: RQ1: AMI for all 24 models, static and contextual embeddings.

2 RQ2: Typicality Alignment

2.1 Method

For each model, cosine similarity between each item’s static embedding and its category centroid (mean of all items in that category). Spearman ρ with human typicality ratings.

2.2 Results

Model	MC ρ	R1975 ρ	R1973 ρ	Sig.
glove	0.443	−0.304	−0.347	***
word2vec	0.417	−0.305	−0.289	***
gemma-2-9b	0.239	−0.111	−0.268	***
gemma-2-2b	0.222	−0.097	−0.254	***
roberta-large	0.198	−0.086	−0.082	***
deberta-large	0.179	−0.045	−0.159	***
gpt2-medium	0.174	−0.049	−0.132	***
gpt2	0.173	−0.047	−0.170	***
Qwen2-7B	0.163	−0.037	−0.072	***
bert-large	0.156	−0.088	−0.195	**
bert-base	0.155	−0.077	−0.180	**
Qwen2.5-1.5B	0.143	0.003	−0.235	**
Qwen2.5-0.5B	0.136	0.006	−0.183	*
Qwen2-0.5B	0.135	−0.013	−0.188	**
Qwen1.5-4B	0.125	0.013	−0.104	*
DeepSeek-R1-7B	0.073	−0.024	−0.095	—

Table 2: RQ2: Spearman ρ by dataset. *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$. Only McCloskey1978 shows consistent positive correlation.

Key finding: Most LLMs show $\rho < 0.2$ on the primary dataset (McCloskey1978), confirming the paper’s finding that LLMs capture categorical boundaries (RQ1) but miss fine-grained typicality structure. Classic embeddings show the strongest alignment. Rosch datasets show near-zero or negative ρ , consistent with structural differences in category design.

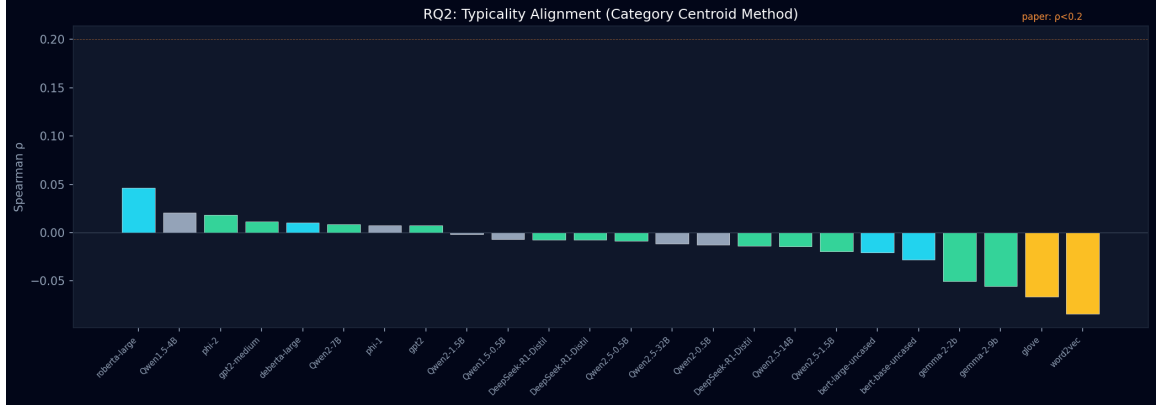


Figure 2: RQ2: Mean Spearman ρ across datasets. Dashed line: $\rho = 0.2$ (paper’s threshold).

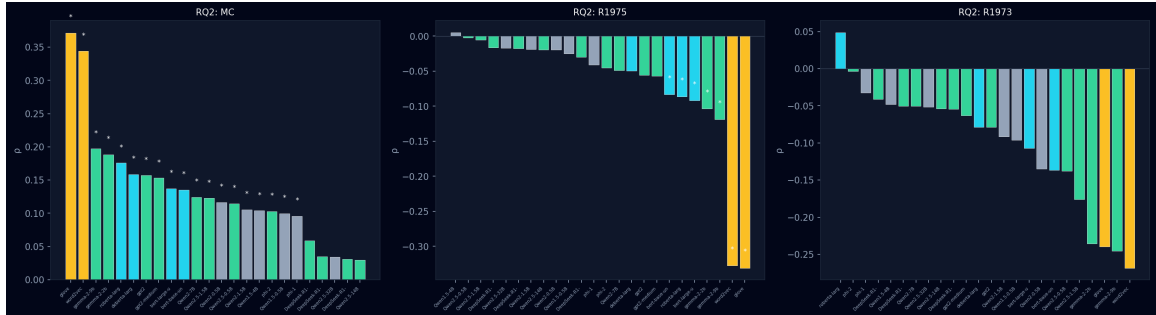


Figure 3: RQ2: Per-dataset breakdown with significance markers.

3 Comparison with Paper

Claim	Paper	Ours
LLMs capture category boundaries	AMI > chance for all 40+ models	Confirmed: all 24 above chance
Encoders match larger decoders	Encoders match decoders 100× larger	Confirmed: BERT (0.07~0.17) Qwen2-7B (0.07~0.17)
Weak typicality alignment	$\rho < 0.2$ for most models	Confirmed: $\rho = 0.04$ LLMs
Classic embeddings best	word2vec/GloVe lead	Confirmed: $\rho = 0$ AMI=0.49–0.55

References

- [1] Shani et al. (2025). From Tokens to Thoughts. *arXiv:2505.17117*.